



e-Pest surveillance architecture

issued to farmers on pest management, enabling them to take preventive steps in time.

The tech challenges

A team of over twelve hardware and software engineers worked continuously for about a year to design and develop the Fieldman ePest surveillance system. Chandrashekar talks about the major challenges that his team encountered and addressed while designing the system hardware: "Incorporating all the modules like GPS, GSM, GPRS and other sensors for data collection along with a touchscreen while meeting the key requirement of making the device rugged enough to operate in extreme environmental conditions was a challenge. On the software front, the major challenge was developing the standard data collection format that would meet the requirements of various stakeholders."

"The other issue was developing a graphical user-interface (GUI) which

would be user-friendly, keeping in mind various operational requirements such as daylight usage, and customising the data for easy field operations, amongst others," he adds.

The prototype was first tested at various locations in Andhra Pradesh and Kenya under FAO's supervision. Based on the feedback and test results, necessary modifications and enhancements were done to the system to enable it to meet the user-requirements. "The system has been designed to ensure low cost of ownership and high return on investment," Chandrashekar affirms.

How the Fieldman works

The agriculture field officer deputed by the agricultural department surveys the field with the Fieldman handheld device having inbuilt sensors and software. Different types of data (including climatic condition, soil data, temperature, humidity, pests and other information beneficial for crop man-

agement) is captured.

This data is transmitted from the field to a central server using GPRS or GSM, or transferred to an Internet-enabled computer through USB cord. The data is stored at the central server and various survey data reports can be generated through its reporting module by picking the data pest-wise, crop-wise and site-wise.

Based on these reports and analysis, agriculture experts can identify the growth of different

pests and diseases across regions on various crops and necessary corrective or preventive measures can be taken.

Going forward

The company is continuously working on adding new features to the system. Chandrashekar divulges: "Based on the feedback from end users, we are continually improving the platform. This is an ongoing process. Currently, our system is Windows-based and we are working on the Android and Linux platforms."

He adds, "Our next version of the Fieldman device will incorporate a digital camera, using which the field officer will be able to take high-resolution pictures of the pests with geo-references. Based on this, we plan to develop a picture-search module, enhance customisation of reports and add 3G support to the device."

"As of now there is no similar solution for pest surveillance and monitoring based on handheld devices," says Chandrashekar. "The Fieldman system will play a crucial role in making significant contribution towards numerous government initiatives for Indian agriculture like Rashtriya Krishi Vikas Yojana, Technology Mission for Development of Horticulture in North-East and National Food Security Mission, which focus on achieving higher productivity and better-quality agriculture produce," he concludes. ●

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The need for pest surveillance

So far the key problems that mar the agricultural sector include lack of standardised methods to gather surveillance data from fields. Another problem is that the field officers appointed by the agricultural departments to report the data from fields often shirk from visiting the fields or report unreliable data. Sometimes the pesticide companies also influence the officers to file an incorrect report to serve their vested interests. As a consequence, agricultural departments are unable to issue reliable advisories to farmers.

Farmers continue to make injudicious use of pesticides, which leads to loss of agricultural yield, increased production cost, and damage to the environment. The agricultural departments are also unable to develop pest-forecasting models and study impact on climate change. Pest surveillance plays an important role in plant-protection and has a direct impact in increasing agricultural yields.